

Mahesh Rajarapu

Doctor of Philosophy (Ph. D)

Department of Computer Science & Engineering

Indian Institute of Technology, Tirupati

Links: [Home Page](#) | [LinkedIn](#) | [GitHub](#) | [Leetcode](#)

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Location: Surat, Gujarat, India

Academic Qualifications

- **PhD in Computer Science and Engineering** (July 2025 – Present)
Indian Institute of Technology Tirupati, Tirupati, Andhra Pradesh, India
- **Integrated Master of Science in Mathematics** (Dec 2020 – June 2025)
Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India
SGPA: 8.13/10.0 | CGPA: 7.55/10.0
- **Board of Intermediate Education** (June 2018 – Feb 2020)
Sri Sarada Educational Institutions, Vijayawada, Andhra Pradesh, India
CGPA: 8.82/10.0
- **Board of Secondary Education** (March 2018)
Bhashyam E.M High School, Bhavanipuram, Andhra Pradesh, India
CGPA: 9.80/10.0

Research Experience

- **Graduate Research Intern** (Jun 2025 - Present)
Indian Institute of Management Mumbai, Mumbai, India | PI: Prof. Manoj K. Tiwari, Director, IIM Mumbai.
 - Developed a novel framework using a fine-tuned Large Language Model (LLM) to optimize drone delivery, integrating real-time meteorological data from the Open-Meteo API for dynamic route planning, energy-efficient battery management, and multi-drone coordination, achieving enhanced reliability and efficiency in urban settings like Delhi.
 - Provided an open-source toolkit with datasets and weather-aware modules to support scalable, sustainable drone logistics.
- **KAI-X Summer Research Internship** (Jun 2024 - Aug 2024)
Korea Advanced Institute of Science and Technology, Daejeon, South Korea | PI: Prof. Chang-Ock Lee.
 - Focused on simulating computed tomography (CT) and developing advanced image reconstruction algorithms, including classical methods like Filtered Back-Projection (FBP) and deep learning techniques.
 - Aimed to enhance image quality and reduce artifacts by integrating deep learning techniques with a deep understanding of CT physics and data acquisition simulation.
- **Summer Internship Program** (May 2024 - Jun 2024)
National Institute of Technology Calicut, Kozhikode, Kerala, India | PI: Dr. Pranesh Das.
 - Developed the Dynamic Batch Size Adjuster (DBSA) algorithm to dynamically optimize batch sizes during training, enhancing convergence rates and stability in deep learning models.
 - Improved model performance and minimized computational complexity by adjusting batch sizes in real-time using mathematical analysis of training dynamics, including loss functions and gradient norms.
- **Research Internship** (Ongoing)
Sardar Vallabhbhai National Institute of Technology, SVNIT, Surat | PI: Dr. Praveen Kumar Chandalia.
 - Working on child face aging using AI to generate synthetic faces through latent space manipulation, with applications in forensics and healthcare.
 - Working on explainable AI algorithms to assess facial image quality, enhancing image processing techniques across various domains.
- **Research Internship Program** (Sept 2023)
Technical University of Munich, Department of Informatics, Virtual – Munich, Germany | PI: Dr. Nassir Navab.
 - Currently working on 3D Ultrasound Compounding for Volume Estimation in Thyroid Diagnostics.
 - Applied PyTorch on Image Computation to develop cutting-edge solutions for improving Multi-model Thyroid Registration.

- **IAS-INSNA-NASI Summer Research Fellowship Program** (May 2023 - July 2023)
Indian Institute of Technology Ropar, DoCSE, Ropar, Punjab, India | PI: Dr. Sudarshan Iyengar.
 - Utilized expertise in machine learning & Deep Learning to work on auto image enhancement, employing advanced techniques and models with optimal gamma values.
 - Applied proficiency in TensorFlow, PyTorch, and JAX to develop cutting-edge solutions for improving image quality and visual appeal.
- **Summer Workout in Mathematics & Programming** (June 2022)
TATA Institute of Fundamental Research - CAM, Bengaluru, Karnataka, India.
 - Studied statistical learning techniques for data analysis, regression, classification, and dimensionality reduction, alongside Python programming proficiency.
 - Developed a solid foundation in real analysis and linear algebra for mathematical and computational problem-solving.

M.Sc. Dissertation

- **Dissertation (Ongoing)** (Expected: 2024-2025)
Sardar Vallabhbhai National Institute of Technology, SVNIT, Surat, | PI: Dr. Shivam Bajpeyi.
 - Currently developing preliminaries on neural network approximation and reconstruction through the exploration of different activation functions. The focus is on understanding how varying activation functions impact the learning process and accuracy of neural network-based models.
 - Investigating advanced methods for reconstructing neural network operators and developing efficient approximations through Gaussian interpolation and other optimization techniques.

Research Publications

- **Application of Gaussian Activated Neural Network in Image Processing** (Submitted)
Mahesh Rajarapu, Shivam Bajpeyi
| Department of Mathematics, Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat-395007.
 - Focuses on developing Gaussian-activated neural network models for improved image processing techniques.
 - Explores the impact of Gaussian interpolation methods on enhancing reconstruction accuracy in image datasets.
- **Dynamic Batch Size Adjuster (DBSA): Enhancing Deep Learning Training Efficiency** (Submitted)
Mahesh Rajarapu, Pranesh Das*, Ebrahim Ghaderpour
| Machine Learning Laboratory, Department of Computer Science and Engineering, National Institute of Technology, Calicut, India.*
 - Proposes a novel adaptive batch size mechanism to enhance deep learning model convergence.
 - Aims to improve training efficiency and stability for various neural network architectures.
- **Applications and Performance Analysis of Neural Network Operators In Image Processing** (Drafting)
Mahesh Rajarapu, Shivam Bajpeyi
| Department of Mathematics, Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India.
 - Investigates the application and performance of neural network operators in enhancing image processing tasks.
 - Analyzes the efficiency and effectiveness of neural network-based methods for image data processing.
- **Hybrid Transformer-GNN Architecture for Multispectral Land Use and Land Cover Classification on EuroSAT** (Drafting)
Mahesh Rajarapu, Ebrahim Ghaderpour*, Pranesh Das
| Department of Earth Sciences - Sapienza University of Rome, Italy.*
 - Proposes a novel hybrid transformer-GNN architecture for EuroSAT LULC classification, combining hierarchical attention and graph-based spectral-spatial modeling.
 - Utilizes 13 multispectral bands to achieve > 99.6% accuracy and superior generalization, outperforming CNN (99.17%) and transformer baselines.

Key Projects

- **CT Simulation and Image Reconstruction Algorithms**
Mentor: Prof. Chang-Ock Lee **Source Code**
 - * Simulating CT data acquisition and implementing reconstruction algorithms like Filtered Back-Projection (FBP).
 - * Exploring deep learning techniques to enhance image quality and reduce artifacts in imaging.
- **Real-time Object Detection with Deep Learning** **Source Code**

- * Developed a real-time object detection system using advanced deep learning techniques.
- * Leveraged YOLOv8 architecture for accurate detection of objects in video streams.
- * Created a solution for real-time visual assistance across various applications.
- **Image CompressNet using Encoder-Decoder Architecture** **Source Code**
 - * Designed an autoencoder architecture for binary image reconstruction using the Keras Functional API.
 - * Developed an encoder-decoder structure with convolutional and upsampling layers to enhance image reconstruction and model performance.
- **Auto Image Enhancement using Optimal Gamma Values** **Source Code**
 - * Applied Power Law Transformations, CLAHE, and Scaling methods for image enhancement.
 - * Developed CNN models for automatic image enhancement, improving contrast, brightness, and saturation.
- **Chronic Kidney Disease Prediction using ML Techniques** **Source Code**
 - * Evaluated various machine learning algorithms to predict kidney disease, identifying the most accurate algorithm.
- **Automated Face Recognition and Attendance Management System** **Source Code**
 - * Developed a face recognition system for entry gate operations.
 - * Implemented a deep learning model for accurate identification and attendance logging.

Technical Skills

- **Programming Skills:** Python, C, SQL, Matlab
- **Deep Learning Frameworks:** TensorFlow, PyTorch, JAX, Flax
- **Machine Learning Frameworks:** Scikit-Learn, XGBoost, LightGBM, Keras
- **Version Control Tools:** Git, GitHub
- **Web Development:** HTML, CSS, JavaScript
- **Other Skills:** $\text{\LaTeX}$, Excel, Power BI, MS Office

Fundings

- **Sponsorship for KAI-X**
 - * Received 1.5 million KRW and housing to participate in the KAI-X Summer Research Internship at Korea Advanced Institute of Science and Technology (KAIST), South Korea (Jun 2024 - Aug 2024).
- **IAS-INSANA-NASI SRFP Fellowship**
 - * Received a monthly stipend for two months during the IAS-INSANA-NASI Summer Research Fellowship Program at IIT Ropar, funded jointly by the Indian Academy of Sciences (IAS), Indian National Science Academy (INSA), and National Academy of Sciences, India (NASI) (May 2023 - July 2023).
- **Personal Funding for Integrated M.Sc.**
 - * Secured complete personal funding for 5 years of Integrated M.Sc. in Mathematics at Sardar Vallabhbhai National Institute of Technology, Surat, covering tuition and academic expenses (Dec 2020 - June 2025, expected).
- **SVNIT Alumni Funding**
 - * Received alumni funding from SVNIT Surat on two occasions to support academic and research endeavors during the Integrated M.Sc. program.

Scholastic Achievements

- Awarded the prestigious **KAI-X Summer Internship 2024** at **Korea Advanced Institute of Science and Technology (KAIST)**, South Korea, selected as one of 16 interns from a competitive global pool.
- Awarded fellowship to work under Dr. Sudarshan Iyengar, Dept. of CSE, IIT Ropar by **Indian Academy of Sciences along with INSA, NASI – Summer Research Fellowship**.
- Secured rank in the merit list: Top 53,000 among 1 million candidates in **NIT-JEE Mains 2020**.
- Secured 7th rank in **National Talent Search Examination (NTSE)** conducted in Andhra Pradesh.
- Secured Zonal 6th rank in **National Mathematics Olympiad**.

External Short Term Courses

- **MaLICO-N**
 - * One Week Online Short Term Course (e-STC) organized by National Institute of Technology, Hamirpur on "Machine Learning and its applications in Information Security, Computer Vision, and Natural Language Processing (MaLICO-N-2023)" during 13-17 March 2023.

– **Advanced and Technical Computing with MATLAB**

- * Participated in the 5 Days Online Faculty Development Programme on Advanced and Technical Computing with MATLAB organized by the Department of Mathematics, School of Engineering, Presidency University, Bengaluru from 25th to 29th July 2023.

Technical Courses/Electives

Data Structures and Algorithms	Artificial Intelligence
Computer Networks	Graph Theory
Ethical Hacking	Natural Language Processing
Block-chain Technology	Computer Vision
Computer Graphics	Database Management Systems

Note: For a comprehensive list of all courses, please click [here](#).

Talks Delivered/Presentations

- About "Low Cost Sanitary Pad Product Manufacturing Machine" in **Indian Institute of Technology, Delhi** ("UNNATI" MAHOTSAV) and this Project/Machine designed under the UNNAT BHARAT ABHIYAN.

Positions of Responsibility

- **Convener, Unnat Bharat Abhiyan**, SVNIT Surat during A.Y. 2022-23.
 - * Inspired by the vision of transformational change in rural development processes by leveraging knowledge institutions to help build the architecture of an Inclusive India.
- **Convener, National Mathematics Day (NMD) 2022**, DoMH, SVNIT Surat.
 - * Organized 11 events at the national level as part of the celebrations of NMD.
- **Head, Media and Designing Committee**, Academic Affairs Council (Student's Council) during A.Y. 2022-23.
 - * Managed media, designed posters, and created videos for all programs conducted under the AAC Student Council.
- **Co-Head, Media and Designing Committee**, General Student Council during 2022-23.
 - * Secured the media, designed posters, and created videos for various programs conducted under the General Student Council.
- **Designer**, Departmental Brochure, Handbook, Pramiti.
 - * Designed and developed the departmental brochure, student handbook, and Pramiti magazine, ensuring high-quality design and content presentation.

References

Dr. Jayesh M. Dhodiya (HoD)
Associate Professor, Department of Mathematics, SVNIT, Gujarat
✉ jmd@amhd.svnit.ac.in

Dr. Shivam Bajpeyi
Assistant Professor, Department of Mathematics, SVNIT, Gujarat
✉ shivambajpeyi@amhd.svnit.ac.in

Dr. Praveen K. Chandaliya
Assistant Professor, Department of Artificial Intelligence, SVNIT
✉ pkc@aid.svnit.ac.in

Languages

Telugu: Bilingual Proficiency

English: Working Proficiency

Hindi: Conversational

"I believe that "Experience" is the word driving our entire world and universe. Every action, movement and endeavor of humanity is ultimately motivated by the desire to gain new experiences."

- Mahesh Rajarapu